

Machine Learning for Cross-Sectional Stock Return Prediction

Course Project Report

Momen Ismail

Machine Learning for Finance, University of Bonn

Instructor: Ivan Gufler

July 27, 2026

Abstract

This paper compares linear, regularized, dimension-reduction and tree-ensemble methods for predicting next-month stock excess returns using a high-dimensional monthly panel of firm characteristics, accounting information and aggregate economic states. The empirical design uses chronological training, validation and test samples, with annual expanding-window folds for hyperparameter selection. Final results are reported after the selected specifications are refitted on the complete development sample and evaluated on the untouched 2020–2025 test period.

Keywords: stock return prediction, machine learning, high-dimensional predictors, out-of-sample evaluation

1 Introduction

Expected stock returns are central to asset pricing, portfolio allocation and risk management. Predicting them is nevertheless difficult because realized monthly returns contain substantial idiosyncratic variation and because predictive relationships may change across firms and market regimes. A model that explains historical observations well may therefore perform poorly when applied to genuinely future data.

Machine-learning methods are potentially useful in this setting because they can process large sets of correlated predictors, impose regularization and represent nonlinearities and interactions. These properties are particularly relevant when the available information includes firm characteristics, accounting variables, market conditions and aggregate economic states. However, greater model flexibility also increases the risk of overfitting. A meaningful empirical comparison must therefore use chronological validation, an untouched test sample and a transparent benchmark.

This paper addresses the following research question:

Can linear, regularized, dimension-reduction and tree-ensemble methods improve next-month stock excess-return forecasts relative to a historical-mean benchmark?

The empirical analysis compares six forecasting approaches under a common information set and evaluation framework: the historical mean, a parsimonious three-variable OLS model, Partial Least Squares, Elastic Net, Random Forest and Gradient Boosting. The high-dimensional specifications use 484 predictors, including stock characteristics, accounting information, market and

VIX variables, Welch–Goyal aggregate states, SIC2 industry controls and characteristic–macro interactions.

The empirical design respects the chronological nature of financial prediction. The initial training period runs from February 1990 through December 2014. Hyperparameters are selected using annual expanding-window validation folds within this period. Fixed and optimized specifications are then compared using the January 2015–December 2019 validation sample. Once the specifications are fixed, the training and validation observations are combined into a development sample, and the models are refitted before being evaluated once on the January 2020–December 2025 test period.

The study is related to the predictive asset-pricing literature that examines whether large sets of firm characteristics can improve expected-return forecasts. In particular, Gu et al. (2020) show how machine-learning methods can accommodate nonlinearities and interactions in high-dimensional asset-pricing data. The analysis also follows the benchmark-oriented perspective of Welch and Goyal (2008), who demonstrate that many apparently sophisticated predictive models struggle to outperform simple historical forecasts outside the estimation sample.

The contribution of the paper is not a new forecasting algorithm. Instead, it provides a transparent and reproducible comparison of models with different levels of complexity under a common timing structure. Particular attention is given to leakage prevention, expanding-window tuning, comparison with the historical-mean benchmark and interpretation of both positive and negative results. This is important because a favorable model ranking does not necessarily imply strong economic predictability or a profitable trading strategy.

The remainder of the paper is organized as follows. ?? describes the data sources, target construction, predictor set and chronological sample design. ?? presents the forecasting models, tuning procedure and evaluation criteria. ?? reports validation and final test results together with model-interpretation evidence. The subsequent section discusses limitations and reproducibility, and the final section concludes.

2 Data

2.1 Sources and observation unit

The observation unit is a stock-month. The stock universe is based on a locked S&P 500-style ticker list that includes both current and historical securities available in the project inputs. The universe is held fixed throughout the production pipeline so that all data-construction and modeling scripts use the same set of ticker identifiers.

Daily stock-price and volume information is obtained from Yahoo Finance. The daily S&P 500 index and VIX series are used to construct aggregate market and volatility variables. Monthly Fama–French data provide the one-month risk-free rate used to calculate excess returns. Annual Compustat data provide accounting characteristics and industry classifications. Aggregate predictive states are obtained from the Welch–Goyal data set.

Daily price histories begin in 1987 so that long-window characteristics, such as 12-month beta and longer-horizon momentum, can be constructed before the start of the modeling sample. After data construction and cleaning, the final monthly panel covers February 1990 through December 2025.

2.2 Target construction

For stock i in month t , the forecasting target is the stock’s excess return in the immediately following calendar month:

$$y_{i,t+1} = r_{i,t+1} - RF_{t+1}, \quad (1)$$

where $r_{i,t+1}$ is the stock return in month $t + 1$ and RF_{t+1} is the corresponding one-month risk-free rate.

The target is constructed only when the next observation for the same ticker belongs to the immediately following calendar month. Observations followed by a gap in the ticker’s monthly history are therefore not treated as valid one-month-ahead targets. This prevents a return measured over several calendar months from being incorrectly interpreted as a one-month return.

The target remains in its original return units. It is not cross-sectionally rank-normalized and is not winsorized in the common model sample. January 2026 price information may be used only to realize the target associated with a December 2025 predictor row. The final set of predictor observations therefore ends in December 2025.

2.3 Information timing and leakage controls

All predictors attached to a row dated month t are constructed using information available by the end of that month. Stock characteristics are calculated from current and historical price information through month t . They do not use the return realized in month $t + 1$.

Annual Compustat observations require special treatment because Compustat `datadate` records the fiscal-year-end date rather than the date on which the accounting information became publicly available. Each annual report is therefore treated as available only six months after its fiscal-year end. For every ticker-month, a backward as-of merge assigns the most recent Compustat report whose assumed availability date is not later than the stock month. Accounting observations that have become excessively stale are not carried forward indefinitely.

Welch–Goyal aggregate states are lagged by one month before entering the predictor set. Consequently, a predictor row dated month t uses the aggregate state recorded for month $t - 1$. This conservative timing rule reduces the possibility that information published only after the end of month t enters the forecast.

The combination of consecutive-month target construction, lagged accounting information and lagged aggregate states ensures that future information is not used in the predictors.

2.4 Missing values and cross-sectional transformation

Missing predictor observations are replaced using the cross-sectional median among firms observed in the same month. This procedure uses no information from future months. When an entire predictor is unavailable for all firms in a particular month, its remaining missing values are set to zero.

Continuous predictors are subsequently transformed into within-month cross-sectional ranks on the interval $[-1, 1]$:

$$x_{i,t}^{\text{rank}} = 2 \frac{\text{rank}(x_{i,t}) - 1}{N_t - 1} - 1, \quad (2)$$

where N_t is the number of firms observed in month t . The transformation preserves the cross-sectional ordering of firms while reducing sensitivity to units, skewness and extreme predictor values.

Binary variables and SIC2 industry indicators remain in their natural indicator form rather than being rank-normalized. The return target also remains untransformed.

2.5 Predictor set and final panel

The final model data contain 484 predictors. These can be divided into 124 base predictors and 360 interaction variables.

The 124 base predictors consist of:

- 45 continuous stock and accounting characteristics;
- 3 binary stock characteristics;
- 4 aggregate market and VIX variables;
- 8 lagged Welch–Goyal state variables;
- 64 SIC2 industry indicators.

The continuous stock and accounting characteristics capture economically important dimensions including:

- size and valuation;
- profitability and cash flow;
- investment and asset growth;
- leverage and liquidity;
- short- and long-horizon momentum;
- return reversals and price trends;
- volatility, downside risk, beta and idiosyncratic volatility;
- trading activity and market liquidity.

To allow the relationship between firm characteristics and future returns to vary with aggregate economic conditions, each of the 45 continuous characteristics is interacted with each of the eight Welch–Goyal state variables:

$$z_{i,t}^{(j,k)} = x_{i,t}^{(j)} \times w_t^{(k)}, \quad (3)$$

where $x_{i,t}^{(j)}$ denotes firm characteristic j and $w_t^{(k)}$ denotes aggregate state k . This produces

$$45 \times 8 = 360$$

characteristic–state interactions.

The final model-ready panel contains 211,059 stock-month observations for 656 tickers. It contains no missing predictor values, infinite values or duplicate ticker-month observations. The large number of predictors and the substantial correlation among characteristics motivate the use of regularization, dimension reduction and nonlinear ensemble methods.

2.6 Chronological sample design

The final panel is divided chronologically into the following samples:

- training sample: February 1990–December 2014;
- validation sample: January 2015–December 2019;
- development sample: February 1990–December 2019;
- test sample: January 2020–December 2025.

Hyperparameters are selected using annual expanding-window folds within the training sample. The first tuning fold estimates the model using observations through December 2004 and evaluates it during 2005. The estimation window is then expanded by one year at a time until the final fold, which estimates the model through December 2013 and evaluates it during 2014.

Fixed and tuned specifications are compared using the separate 2015–2019 validation sample. After the final specifications have been selected, the training and validation observations are combined into the development sample. Each model is then refitted on the complete development period and evaluated once on the 2020–2025 test sample.

Random cross-validation is not used because it would mix earlier and later market observations and allow future conditions to influence model selection. The expanding-window design instead approximates the information sequence faced by an investor who repeatedly estimates a model using only data available at the forecast date.

3 Methodology

3.1 Forecasting models

The general forecasting problem is

$$y_{i,t+1} = f(\mathbf{x}_{i,t}) + \varepsilon_{i,t+1}, \quad (4)$$

where the function $f(\cdot)$ differs across models.

Historical Mean. The benchmark predicts the mean target observed in the permitted estimation sample. It ignores firm characteristics and represents the performance of a simple constant forecast.

OLS-3. The parsimonious OLS specification uses size, book-to-market and 12-month momentum. These variables offer an interpretable reference between the constant benchmark and the high-dimensional models.

Partial Least Squares. PLS constructs supervised latent components from the 499 predictors. Unlike unsupervised dimension reduction, the components are chosen to be related to the target. PLS is particularly relevant when predictors are numerous and strongly correlated (Wold et al.

1984).

Elastic Net. Elastic Net combines L_1 and L_2 regularization:

$$\hat{\beta} = \arg \min_{\beta} \left[\frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \alpha \left(\rho \|\beta\|_1 + \frac{1-\rho}{2} \|\beta\|_2^2 \right) \right]. \quad (5)$$

The L_1 component permits variable selection, while the L_2 component stabilizes estimates when predictors are correlated (Tibshirani 1996; Zou and Hastie 2005).

Decision Tree. A regression tree recursively divides the predictor space and assigns a constant prediction to each terminal leaf. It can represent threshold effects and interactions but is sensitive to changes in the estimation sample.

Random Forest. Random Forest averages predictions across bootstrap trees and considers a random subset of predictors at each split (Breiman 2001). Averaging is intended to reduce the variance of an individual tree while preserving nonlinear predictive relationships.

3.2 Evaluation metrics

The primary criterion is monthly mean squared error:

$$\text{MSE}_{\text{month}} = \frac{1}{T} \sum_{t=1}^T \left[\frac{1}{N_t} \sum_{i=1}^{N_t} (y_{i,t+1} - \hat{y}_{i,t+1})^2 \right]. \quad (6)$$

Errors are first averaged across firms and then across months, preventing months with more observations from receiving greater weight.

The report also presents RMSE, MAE, prediction-target correlation and out-of-sample R^2 :

$$R_{\text{OOS}}^2 = 1 - \frac{\sum_{i,t} (y_{i,t+1} - \hat{y}_{i,t+1})^2}{\sum_{i,t} (y_{i,t+1} - \hat{y}_{i,t+1}^{\text{HM}})^2}. \quad (7)$$

A positive value means that the model reduces squared forecast error relative to the historical mean. A negative value means that the model performs worse than the benchmark.

4 Preliminary Results

4.1 Overall predictive performance

Table 1: Preliminary test-period performance, 2020–2025

Rank	Model	MSE	RMSE	MAE	R_{OOS}^2	Correlation
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Notes: Models are ranked by monthly MSE. Correlation is undefined for constant or near-constant predictions. Positive R_{OOS}^2 indicates an improvement over the historical-mean benchmark.

PLS provides the lowest monthly MSE and the highest out-of-sample R^2 . Its R_{OOS}^2 of 0.00179 means that it reduces squared forecast error by approximately 0.18% relative to the historical mean. The result is positive, but economically small.

Random Forest ranks second with an out-of-sample R^2 of approximately 0.058%. OLS-3 ranks third with approximately 0.049%. The difference between these two models is minimal, indicating that the large increase in model complexity delivered by Random Forest produces only a small additional reduction in squared error relative to the parsimonious OLS specification.

Elastic Net and the historical mean produce almost identical results. The optimized Elastic Net contains no coefficients above the specified tolerance and therefore behaves as a constant predictor. This is a meaningful negative finding: after chronological validation, the regularized model found no sufficiently stable incremental signal among the 499 predictors.

The Decision Tree performs considerably worse than all other models. Its out-of-sample R^2 is approximately -13.3% , and its prediction-target correlation is negative. The individual tree appears to fit unstable splits that do not generalize to the test sample.



Figure 1: Preliminary test-period out-of-sample R^2 by model

?? emphasizes both the ranking and the limited absolute predictive power. PLS, Random Forest and OLS-3 outperform the benchmark, but all positive values remain close to zero. The Decision Tree is qualitatively different because its additional flexibility substantially increases forecast error.

The MAE ranking differs slightly from the MSE ranking. The historical mean and Elastic Net have the lowest MAE, while PLS has the lowest MSE. This suggests that PLS reduces some relatively large forecast errors, which receive more weight under squared loss, without reducing the typical absolute error.

4.2 Financial interpretation

The small positive out-of-sample R^2 values represent small reductions in forecast error relative to the historical-mean benchmark. They provide evidence of weak statistical predictability in the current project version, but they do not establish profitable trading. In particular, MSE

does not account for portfolio turnover, transaction costs, liquidity, short-selling constraints or risk adjustment. Nor does it show that the cross-sectional ordering of forecasts can be converted into stable implementable positions. The present evidence should therefore be interpreted as a preliminary comparison of forecasting methods, not as a ready-to-use investment strategy.

4.3 Performance across test years

The preliminary pooled ranking does not imply stable performance in every year. PLS and Random Forest achieve positive yearly out-of-sample R^2 in half of the six test years. OLS-3 does so in one third of the years. The Decision Tree beats the benchmark in only one year.



Figure 2: Out-of-sample R^2 by model and test year

The yearly results demonstrate that even the best overall model does not consistently outperform the historical mean. This instability is important because the test period contains only six years and includes unusually different market environments. The overall advantage of PLS should therefore be interpreted as an average test-period result rather than evidence of stable performance in every regime.

4.4 Forecast dispersion



Figure 3: Distribution of preliminary test-period predictions

The prediction distributions help explain the model results. Elastic Net is absent as a meaningful distribution because it produces constant or near-constant forecasts. PLS and OLS generate relatively concentrated predictions, while the tree-based models produce more dispersed forecasts. Dispersion alone does not indicate predictive quality. The Decision Tree produces variable forecasts but performs poorly because those forecasts are systematically misaligned with realized returns.

4.5 Feature importance



Figure 4: Largest PLS coefficients by absolute magnitude

PLS coefficients measure conditional predictive associations after preprocessing and scaling. Their magnitude should not be interpreted as a causal effect or as an estimated risk premium. Because PLS combines information through latent components, the coefficients reflect the joint high-dimensional prediction problem rather than isolated univariate relationships.



Figure 5: Top Random Forest predictors based on impurity importance

Random Forest importance records how strongly variables reduce squared error inside the fitted trees. It is useful for describing where the model obtains its predictive variation, but it has important limitations. Correlated variables can divide importance among themselves, and impurity importance can favor predictors with more available split points. The results should therefore be interpreted at the level of broad predictor groups rather than as a precise ranking of causal variables.

5 Critical Reflection

5.1 Predictive gains are small

The main limitation is the weak absolute predictive performance. Although PLS ranks first, its out-of-sample R^2 is only approximately 0.18%. Random Forest and OLS-3 improve on the benchmark by even less. Most variation in monthly stock returns therefore remains unexplained.

A ranking can still be statistically informative, but it should not be confused with strong forecasting ability. The results do not establish that the small differences between the leading models would remain stable in another period.

5.2 Flexible models do not automatically perform better

The results provide two important failures. First, the individual Decision Tree performs much worse than the historical mean. Its threshold splits are apparently too unstable for noisy monthly return data. Random Forest improves substantially by averaging across trees, illustrating the benefit of variance reduction.

Second, Elastic Net selects no nonzero coefficients. This does not imply a software error. Given the selected tuning parameters and chronological validation procedure, the model preferred complete shrinkage over unstable predictor effects. The result suggests that individual linear signals are weak relative to the noise in the target.

5.3 Implementation issues identified during the project audit

The project audit identified six implementation and reproducibility problems that prevent the current outputs from being described as final results:

- The ticker-level target lead contains 26 nonconsecutive transitions in which the next available observation is not the immediately following calendar month.
- Welch–Goyal macro variables are merged in the same month rather than with the intended explicit one-month availability lag.
- The documentation and implementation are inconsistent about the tuning periods and the model-selection rule.
- The top-level `run_pipeline.sh` became stale after the model code was reorganized and references stages that no longer match the active workflow.
- The output structure contains excessive duplicate and legacy artifacts, which makes it harder to identify the authoritative results.
- Software dependencies are not pinned to exact versions, limiting exact reproducibility across environments.

These issues do not make the current tables useless: they remain informative about model behavior and expose several robust negative findings. They do mean, however, that the numerical rankings must be treated as preliminary. Before the complete rebuild, the target continuity rule and macro lag will be corrected, the tuning design and documentation will be reconciled, the pipeline entry point will be repaired, dependencies will be pinned, and authoritative outputs will be separated from legacy artifacts. The existing output folder will then be deleted and regenerated from the permanent inputs in one clean chronological run.

5.4 Data limitations

The stock universe is not a fully point-in-time CRSP constituent universe. Ticker-based matching may be affected by symbol changes, mergers and incomplete historical membership records. Survivorship and universe-selection bias cannot be ruled out completely.

Yahoo Finance does not provide the security master, delisting returns and microstructure information available in CRSP. Adjusted price histories may also be revised. Compustat fundamentals are lagged by six months, but exact public filing dates are not used. The resulting timing design is conservative, though not perfectly point in time.

The use of annual Compustat information also means that accounting variables change less frequently than market variables. Some predictors are proxies for variables used in the published asset-pricing literature and may not reproduce their original definitions exactly.

5.5 Validation and specification risk

Annual expanding-window validation respects chronology and reduces direct leakage. Nevertheless, the project contains many researcher choices: predictor construction, interaction design, preprocessing, tuning grids and sample dates. These decisions may adapt indirectly to known characteristics of the data.

An untouched holdout period remains credible only if no model changes are made after inspecting its performance. The current 2020–2025 outputs have already been examined and are therefore reported only as preliminary evidence. The corrected workflow must be specified before the clean rebuild, and any subsequent specification search would require a genuinely new holdout period.

5.6 No economic trading evaluation

The project evaluates prediction error but does not construct portfolios. Consequently, it does not measure turnover, transaction costs, market impact, short-sale constraints or risk-adjusted investment returns. A small reduction in MSE cannot automatically be interpreted as an economically profitable strategy.

A useful extension would specify a portfolio rule before opening a new test sample. The evaluation could then report gross and net returns, volatility, Sharpe ratios, turnover and maximum drawdown. That analysis would have to be kept separate from the current test period to avoid using the same observations for model assessment and strategy design.

5.7 What I would do differently

With improved data access and additional time, the study could be strengthened in four ways:

- use a point-in-time CRSP/Compustat universe with permanent identifiers and delisting returns;
- use exact filing dates for accounting information;
- retain a longer untouched test period or evaluate several historical forecast origins;
- conduct a pre-specified portfolio evaluation with realistic trading costs.

Interpretability could also be improved by examining feature groups, repeated time-block importance and stability across different expanding windows rather than relying on a single definitive importance ranking.

6 Conclusion

This preliminary report examines whether machine-learning methods improve next-month stock excess-return prediction relative to a historical-mean benchmark. The models are selected chronologically and evaluated on a held-out 2020–2025 test period in the current implementation.

PLS performs best according to monthly mean squared error and produces an out-of-sample R^2 of approximately 0.18%. Random Forest and OLS-3 also slightly outperform the benchmark. Elastic Net selects no predictors and effectively collapses to the historical mean. The Decision

Tree performs substantially worse, showing that flexibility without sufficient variance control can be harmful.

The research question can therefore be answered cautiously. Some statistical and machine-learning methods improve on the historical mean, but the gains are small and unstable across years. The results do not demonstrate strong return predictability or a profitable investment strategy. Their principal lesson is that, in noisy financial prediction problems, disciplined chronology, simple benchmarks and honest reporting of failures are at least as important as model complexity. Because the audit identified target-continuity, macro-timing and workflow inconsistencies, these results are not the final project evidence. Those issues will be corrected before the output folder is deleted and the project is rebuilt cleanly from its permanent inputs.

References

- Breiman, L. (2001). “Random Forests”. In: *Machine Learning* 45, pp. 5–32.
- Gu, S., B. Kelly, and D. Xiu (2020). “Empirical Asset Pricing via Machine Learning”. In: *The Review of Financial Studies* 33.5, pp. 2223–2273.
- Tibshirani, R. (1996). “Regression Shrinkage and Selection via the Lasso”. In: *Journal of the Royal Statistical Society: Series B* 58.1, pp. 267–288.
- Welch, I. and A. Goyal (2008). “A Comprehensive Look at the Empirical Performance of Equity Premium Prediction”. In: *The Review of Financial Studies* 21.4, pp. 1455–1508.
- Wold, S. et al. (1984). “The Collinearity Problem in Linear Regression: The Partial Least Squares Approach to Generalized Inverses”. In: *SIAM Journal on Scientific and Statistical Computing* 5.3, pp. 735–743.
- Zou, H. and T. Hastie (2005). “Regularization and Variable Selection via the Elastic Net”. In: *Journal of the Royal Statistical Society: Series B* 67.2, pp. 301–320.

A Code and Reproducibility Overview

The repository implements the empirical workflow as a sequence of auditable stages rather than embedding source code in this report:

1. acquisition of permanent external inputs and cleaning of daily stock histories;
2. construction of monthly stock features and the next-month target;
3. merging of lagged fundamentals and aggregate macroeconomic variables;
4. final missing-value treatment and within-month rank normalization;
5. chronological hyperparameter tuning with expanding estimation windows;
6. preliminary test-period refitting and evaluation;
7. interpretation, figure generation and report production.

The complete Python and shell code is provided in the accompanying repository. This document reads its principal tables and figures from the existing interpretation outputs and does not retrain models or regenerate predictions during compilation. Before the final clean rebuild, the audited timing, workflow and dependency issues described in ?? will be corrected.

B Additional Results

B.1 Model complexity

Table 2: Current model complexity

Model	Predictors	Nonzero	PLS comp.	Tree depth	Forest trees

B.2 Fixed versus optimized specifications

Table 3: Validation performance of fixed and optimized specifications

Model family	Fixed MSE	Optimized MSE	MSE improvement	ΔR^2_{OOS}

B.3 Additional tree interpretation



Figure 6: Top Decision Tree predictors based on impurity importance

The Decision Tree importance figure should be interpreted cautiously because the model has strongly negative out-of-sample performance. A variable can be important to a poorly performing fitted model without containing stable predictive information.